

Seminar 6. Numerical characteristic of random variables

X discrete with p.d.f. $\left(\begin{matrix} x_i \\ p_i \end{matrix} \right)_{i \in \mathbb{N}}$

then $E(X) = \sum_{i=1}^{\infty} p_i \cdot x_i$

$$E(X^k) = \sum_{i=1}^{\infty} p_i \cdot x_i^k$$

X continuous, with p.d.f. $f_X : \mathbb{R} \rightarrow \mathbb{R}$

then $E(X) = \int_{\mathbb{R}} x \cdot f_X(x) dx$

$$E(X^k) = \int_{\mathbb{R}} x^k \cdot f_X(x) dx$$

$$V(X) = E((X - E(X))^2) = E(X^2) - E(X)^2$$

$$\sigma_X = \sqrt{V(X)}$$

• $E(\alpha \cdot X + \beta \cdot Y) = \alpha \cdot E(X) + \beta \cdot E(Y)$

• if X, Y independent $\Rightarrow V(X+Y) = V(X) + V(Y)$

• $V(\alpha \cdot X) = \alpha^2 V(X)$

• $V(\alpha X + \beta) = \alpha^2 V(X)$

• $\text{cov}(X, Y) = E((X - E(X))(Y - E(Y))) = E(XY) - E(X) \cdot E(Y)$

• $\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$ correlation coefficient

↳ if X, Y independent $\Rightarrow \rho_{X,Y} = 0 \Rightarrow X, Y$ uncorrelated

but X, Y uncorrelated $\nRightarrow X, Y$ independent

1. Every day, the number of network blackouts has the following pdf

$$X \begin{pmatrix} 0 & 1 & 2 \\ 0.7 & 0.2 & 0.1 \end{pmatrix}.$$

A small internet trading company estimates that each network blackout costs them \$500.

- a) How much money can the company expect to lose each day because of network blackouts?
b) What is the standard deviation of the company's daily loss due to blackouts?

a) $Y = \text{the daily sum} = 500 \cdot X$

$$E(Y) = E(500 \cdot X) = 500 \cdot E(X)$$

$$E(X) = 0 \cdot 0.7 + 1 \cdot 0.2 + 2 \cdot 0.1 = 0.4$$

$$\Rightarrow E(Y) = 500 \cdot 0.4 = 200 \$$$

b) $V(Y) = V(500 \cdot X) = 500^2 \cdot V(X)$

$$V(X) = E(X^2) - E(X)^2$$

$$X^2 \begin{pmatrix} 0 & 1 & 4 \\ 0.7 & 0.2 & 0.1 \end{pmatrix}$$

$$= (0 \cdot 0.7 + 1 \cdot 0.2 + 4 \cdot 0.1) - 0.16 = 0.6 - 0.16 = 0.44$$

$$\Rightarrow V(Y) = 500^2 \cdot 0.44$$

$$\Rightarrow \sigma_Y = 500 \sqrt{0.44} \approx 331 \$$$

2. (Refer to Problem 5 in Sem. 3) About ten percent of computer users in a public library do not close Windows properly. On the average, how many users *do* close Windows properly before someone *does not*?

Sol.: $X = \# \text{ of users that close W properly before someone doesn't}$
 $= \# \text{ of failures before the 1st success}$

Success = not closing Windows properly

$$\Rightarrow p = P(\text{"success"}) = 0.1$$

$$\Rightarrow X \sim \text{Geo}(0.1)$$

$$3) \quad X \sim \text{Geo}(p) \Rightarrow X \left(p q^k \right) \leq 1$$

where $q = 1 - p$

$$E(X) = \sum_{k=0}^{\infty} k \cdot p q^k = p \cdot \sum_{k=0}^{\infty} k q^k$$

$$\begin{aligned} f(x) &= 1 + x + x^2 + \dots + x^n + \dots = \\ &= \lim_{n \rightarrow \infty} (1 + x + x^2 + \dots + x^n) = \lim_{n \rightarrow \infty} \frac{x^{n+1} - 1}{x - 1} \quad \text{if } |x| < 1 \\ &= \frac{1}{1-x} \end{aligned}$$

Apply $\frac{d}{dx}$ and get:

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots + n x^{n-1} + \dots$$

We can then deduce that:

$$E(X) = p \cdot q \cdot \sum_{k=0}^{\infty} k \cdot q^{k-1} = p \cdot q \cdot \left. \frac{1}{(1-x)^2} \right|_{x=q}$$
$$= p \cdot q \cdot \frac{1}{(1-q)^2} = \frac{p \cdot q}{p \cdot q} = \frac{q}{p}$$

In our case $E(X) = \frac{0.9}{0.1} = 9$

3. (Refer to Problem 1 in Sem. 5) The lifetime, in years, of some electronic component is a random variable with density

$$f(x) = \begin{cases} \frac{3}{x^4}, & \text{for } x \geq 1 \\ 0, & \text{for } x < 1. \end{cases}$$

How many years, on the average, can we expect that electronic equipment to last?

$$\begin{aligned} \text{Sol. : } E(X) &= \int_{\mathbb{R}} x \cdot f(x) dx = \int_1^{\infty} x \cdot \frac{3}{x^4} dx = \\ &= \int_1^{\infty} \frac{3}{x^3} dx = \frac{3}{-2} x^{-2} \Big|_1^{\infty} = -\frac{3}{2} \left(\frac{2}{x^2} \Big|_1^{\infty} \right) = \\ &= \frac{3}{2} \end{aligned}$$

4. (Optimal portfolio) A businessman wants to invest \$600 and has two companies to choose from, company A, where shares cost \$20 each and company B, where shares cost \$30 per share. The market analysis shows that for company A the return per share is distributed as follows: lose \$1 with probability 0.2, win \$2 with probability 0.6, or win/lose nothing. For company B: lose \$1 with probability 0.3, win \$3 with probability 0.6, or win/lose nothing. The returns from the two companies are independent. In order to maximize the expected return and minimize the risk, which way is better to invest:

- a) all money in company A;
- b) all money in company B;
- c) half the amount in each?

$$A \begin{pmatrix} -1 & 0 & 2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$$

$$a) S_1 = 30 A$$

$$E(S_1) = 30 \cdot E(A)$$

$$E(A) = -1 \cdot 0.2 + 0 \cdot 0.2 + 2 \cdot 0.6 = 1$$

$$\Rightarrow E(S_1) = 30$$

$$V(S_1) = V(30A) = 30^2 \cdot V(A)$$

$$A^2 \begin{pmatrix} 1 & 0 & 4 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$$

$$V(A) = E(A^2) - E(A)^2 = (1 \cdot 0.2 + 4 \cdot 0.6) - 1 = 0.2 + 2.4 - 1 = 1.6$$

$$\Rightarrow V(S_1) = 30^2 \cdot 1.6 = 900 \cdot 1.6 = 1440$$

$$b) S_2 = 20 \cdot B \Rightarrow E(S_2) = 20 \cdot E(B)$$

$$B \begin{pmatrix} -1 & 0 & 3 \\ 0.3 & 0.7 & 0.6 \end{pmatrix}$$

$$E(B) = -1 \cdot 0.3 + 3 \cdot 0.6 = 1.5$$

$$E(S_2) = 20 \cdot 1.5 = 30$$

$$V(B) = V(20 - B) = 20^2 \cdot V(B)$$

$$V(B) = E(B^2) - E(B)^2 = \begin{matrix} B \\ \begin{pmatrix} 1 & 0 & 5 \\ 0.3 & 0.1 & 0.6 \end{pmatrix} \end{matrix}$$

$$= 0.3 + 5.4 - 2.25$$

$$= 5.7 - 2.25 = 3.45$$

$$V(S_2) = 20^2 \cdot 3.45 = 400 \cdot 3.45 = 1380$$

$$(c) \quad S_3 = 15A + 10B$$

$$E(S_3) = E(15A + 10B) = 15E(A) + 10E(B) =$$

$$= 15 \cdot 1 + 10 \cdot 1.5 = 15 + 15 = 30$$

$$V(S_3) = V(15A + 10B) =$$

A, B independent

$$= V(15A) + V(10B) =$$

$$= 15^2 V(A) + 10^2 V(B) =$$

$$= 225 \cdot 1.6 + 100 \cdot 3.45$$

$$= 300 + 345 = 705$$

$\Rightarrow$ Same expectation, < 40 ,
lowest variance, so C is the
winner

Let $p \in (0, 1)$

Consider the strategy:

$$S_p = p \cdot 30 \cdot A + (1-p) \cdot 20 \cdot B$$

Find p so that $E(S_p)$ is maximized
and $V(S_p)$ is minimized

$$E(S_p) = 30p + 20(1-p) \cdot 1.5$$

$$= 30p + 30(1-p) = 30$$

$$V(S_p) = 900p^2 \cdot 1.6 + 400(1-p)^2 - 3,45$$

$$= 1440p^2 + 1440(1-p)^2$$

$$= 1440 (2p^2 - 2p + 1)$$

$$\min(2p^2 - 2p + 1) = -\frac{4-8}{8} = \frac{1}{2}$$

5. (Reduced Variables). Let X be a random variable with mean $E(X)$ and standard deviation $\sigma(X) = \sqrt{V(X)}$. Find the mean and variance of $Y = \frac{X - E(X)}{\sigma(X)}$.

$$\begin{aligned} E(Y) &= E\left(\frac{X - E(X)}{\sigma(X)}\right) = \frac{1}{\sigma(X)} E(X - E(X)) = \\ &= \frac{1}{\sigma(X)} \cdot \left(E(X) - \underbrace{E(E(X))}_{=E(X)}\right) = 0 \end{aligned}$$

$$\begin{aligned} V(Y) &= V\left(\frac{X - E(X)}{\sigma(X)}\right) = \frac{1}{\sigma(X)^2} \cdot V(X - E(X)) = \\ &= \frac{1}{\sigma(X)^2} \cdot V(X) = 1 \end{aligned}$$

6. The joint density function of the vector (X, Y) is $f(x, y) = x + y$, $(x, y) \in [0, 1] \times [0, 1]$. Find

a) the means and variances of X and Y ;

b) the correlation coefficient $\rho(X, Y)$.

$$a) f_X(x) = \int_0^1 f(x, y) dy$$

$$\begin{aligned} E(X) &= \int_0^1 x \cdot f_X(x) dx = \int_0^1 \int_0^1 x \cdot f(x, y) dx dy = \\ &= \int \int_{[0,1] \times [0,1]} (x^2 + xy) dx dy = \int_0^1 \left(x^2 y + \frac{1}{2} xy^2 \right) \Big|_0^1 dy = \\ &= \int_0^1 \left(x^2 + \frac{1}{2} x \right) dx = \frac{1}{3} x^3 + \frac{1}{4} x^2 \Big|_0^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12} \end{aligned}$$

$$\Rightarrow E(X) = \frac{7}{12}$$

$$V(X) = E(X^2) - E(X)^2$$

$$E(X^2) = \int_0^1 \int_0^1 x^2 \cdot f(x, y) dx dy =$$

$$= \int \int_{[0,1] \times [0,1]} (x^3 + x^2 y) dx = \int_0^1 \left(x^3 y + \frac{x^2}{2} y^2 \right) \Big|_0^1 dy =$$

$$= \int_0^1 \left(x^3 + \frac{x^2}{2} \right) dx = \frac{1}{4} x^4 + \frac{x^3}{6} \Big|_0^1 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

$$\Rightarrow V(X) = \frac{5}{12} - \frac{49}{144} = \frac{11}{144}$$

$$\Rightarrow V(Y) = \frac{11}{144} \Rightarrow \sigma_X = \sigma_Y = \frac{\sqrt{11}}{12}$$

$$\rho_{X,Y} = \frac{E(X,Y) - E(X) \cdot E(Y)}{\sigma_X \cdot \sigma_Y}$$

$$E(X,Y) = \int_0^1 \int_0^1 x \cdot y \cdot (x+y) \, dx \, dy$$

$$= \iint_{(0,1)^2} (x^2 y + x y^2) \, dx \, dy =$$

$$= \int_0^1 \left(\frac{x^2 y^2}{2} + \frac{x y^3}{3} \right) \Big|_0^1 \, dy =$$

$$= \int_0^1 \left(\frac{x^2}{2} + \frac{x}{3} \right) \, dx = \frac{2}{6} x^3 + \frac{1}{6} x^2 \Big|_0^1 = \frac{2}{6} + \frac{1}{6} = \frac{1}{3}$$

$$\Rightarrow \rho_{X,Y} = \frac{\frac{1}{3} - \frac{49}{144}}{\frac{11}{144}} = \frac{48 - 49}{11} = -\frac{1}{11}$$

$\Rightarrow X, Y$ are weakly inversely correlation

7. Let X be a discrete random variable with pdf $X \begin{pmatrix} -1 & 0 & 1 \\ \sin^2 a & \cos 2a & \sin^2 a \end{pmatrix}$, $a \in (0, \frac{\pi}{4})$. For any $k \in \mathbb{N}^*$, let $Y_k = X^{2k-1}$ and $Z_k = X^{2k}$. Find $\rho(Y_k, Z_k)$. (In particular, X and X^2 are uncorrelated, but *not* independent).

Sol.: $\rho(Y_k, Z_k) = \frac{\text{cov}(Y_k, Z_k)}{\sigma_{Y_k} \cdot \sigma_{Z_k}} = \frac{E(Y_k Z_k) - E(Y_k) \cdot E(Z_k)}{\sigma_{Y_k} \cdot \sigma_{Z_k}}$

$$Y_k = \begin{pmatrix} -1 & 0 & 1 \\ \sin^2 a & \cos 2a & \sin^2 a \end{pmatrix} = X$$

$$E(Y_k) = -1 \cdot \sin^2 a + 1 \cdot \sin^2 a = 0$$

$$Z_k = \begin{pmatrix} 0 & 1 \\ \cos 2a & 2 \sin^2 a \end{pmatrix}$$

$$E(Z_k) = 2 \sin^2 a$$

$$Y_k Z_k = \begin{pmatrix} -1 & 0 & 1 \\ 2 \sin^4 a & 1 - 4 \sin^4 a & 2 \sin^4 a \end{pmatrix}$$

$$E(Y_k Z_k) = -2 \sin^4 a + 2 \sin^4 a = 0$$

$$\Rightarrow \text{cov}(Y_k, Z_k) = E(Y_k Z_k) - E(Y_k) \cdot E(Z_k) = 0 - 0 \cdot 2 \sin^2 a = 0$$

$$\Rightarrow \rho_{Y_k, Z_k} = 0 \Rightarrow Y_k, Z_k \text{ uncorrelated}$$

$$P(Y_k = -1, Z_k = 1) = P(X = -1) = \sin^2 a$$

$$P(Y_k = -1) = \sin^2 a \quad P(Z_k = 1) = 2 \sin^2 a$$

2) If $\tilde{r}_2$ and z_2 were independent, then we would have

$$2\sin^2 a = \sin^2 a - 2\sin^2 a$$

so we would have $\sin^2 a (\sin^2 a - 1) = 0$
which is not true if $a \neq 0, \pi/2$, which is the case

$\Rightarrow \tilde{r}_2$ and z_2 are not independent